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SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Concept Mapping
Weightage:	25%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking.		
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Section / Batch:	Btech AI4 DS	Date:	20/7/26

Instructions to Students

- Identify the key computer architecture concepts, components, and performance parameters mentioned in the given topic or scenario.
- Organize the identified concepts logically by establishing relationships between CPU, memory, I/O devices, bus structures, instruction execution cycle, and performance metrics.
- Represent the concepts using an appropriate concept map with clear nodes, connecting lines, and meaningful linking phrases.
- Demonstrate the hierarchy and interdependence among concepts, showing how one concept influences or interacts with another.
- Ensure the concept map is complete, accurate, well-structured, and easy to understand by using appropriate terminology and logical connections.

QUESTIONS

- A computer manufacturer is designing a workstation for engineering simulations that frequently exchanges data between the processor, memory, and peripheral devices. Design a concept map illustrating the interaction among the CPU, Memory Unit, Input Unit, Output Unit, and I/O devices, showing how instructions and data move through the system during program execution. How do the functional units of a computer system cooperate to execute instructions efficiently?
- A software developer notices that an application takes longer to execute because several instructions require multiple memory accesses. Develop a concept map showing how the Fetch, Decode, and Execute stages of the instruction cycle are related to memory access, instruction execution, and control flow. How does each stage of the instruction execution cycle contribute to the successful execution of a program?
- A data center experiences communication delays because the processor, memory, and I/O devices frequently compete for access to the system bus. Prepare a concept map demonstrating the relationships among Single-bus, Multi-bus, and Hierarchical Bus organizations and their connections to the CPU, Memory, and I/O devices. How do different bus organizations improve communication efficiency and reduce system bottlenecks?
- A microprocessor programmer is developing software for both 8085 and 8086 systems and must understand how instructions access data stored in memory. Represent the relationships among the instruction sets, register organization, addressing modes, memory organization, and memory segmentation (8086) using a concept map. How do the architectural differences between the 8085 and 8086 affect instruction execution and memory management?
- A firmware engineer is examining machine code stored in memory while debugging an embedded application. Create a concept map explaining the relationships between Binary and Hexadecimal Number Systems, instruction representation, memory storage, data conversion, and CPU interpretation. How do binary and hexadecimal representations simplify instruction execution, memory organization, and debugging in computer systems?

Code of Conduct

I, _____ (Name) Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

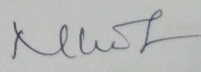
RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2-3 Marks)	Level 1 – Beginning (0-1 Mark)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

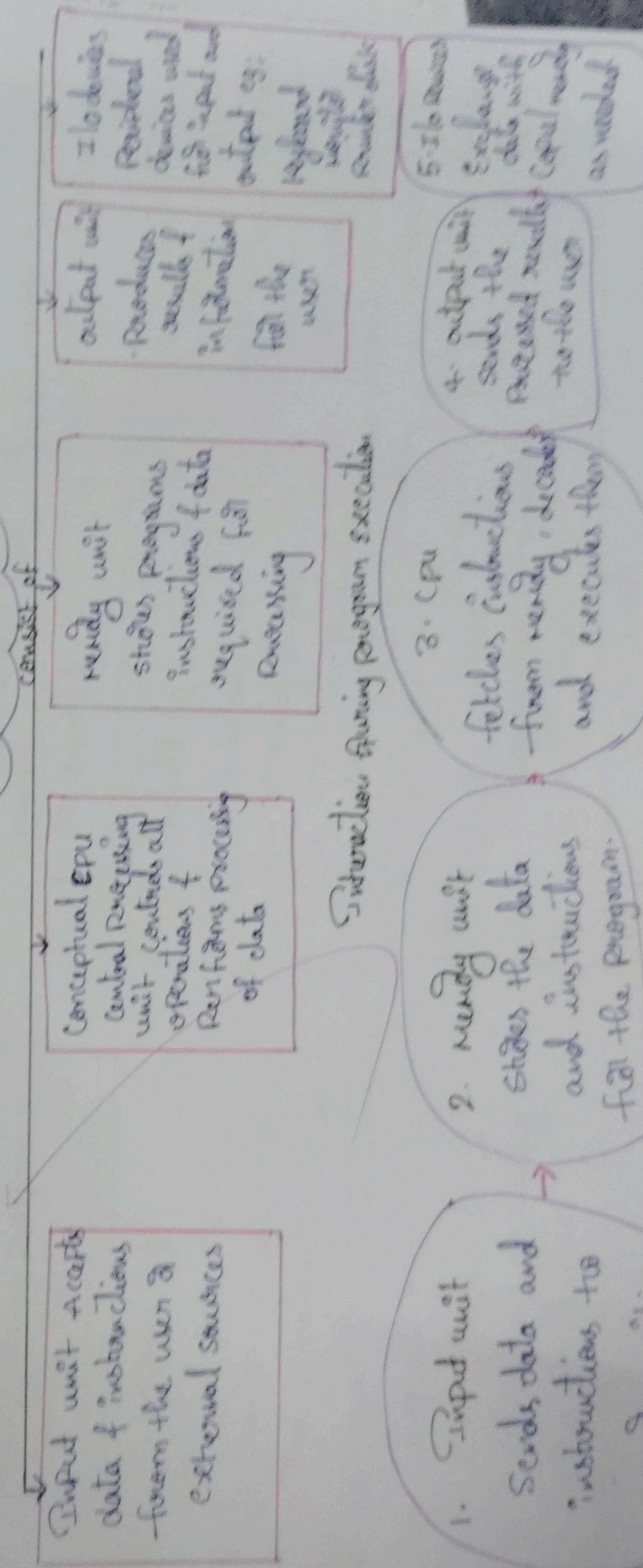
Q. No	Concept Identification (25)	Linkages (25)	Organization (20)	Integration of Literature (20)	Clarity (10)	Total (out of 100)
1	4 / 4	3 / 4	3 / 4	3 / 4	3 / 4	
2	3 / 4	3 / 4	3 / 4	3 / 4	4 / 4	
3	3 / 4	3 / 4	3 / 4	3 / 4	4 / 4	
4	4 / 4	4 / 4	3 / 4	3 / 4	3 / 4	
5	3 / 4	3 / 4	3 / 4	3 / 4	3 / 4	
	21 / 25	20 / 25	15 / 20	15 / 20	9 / 10	80 / 100

Course Coordinator


 Faculty Incharge

1. Functional units of computer system.

Computer system
A combination of components
that work together to execute
Program.



2. Instruction execution cycle.

Instruction execution cycle decodes & executes instructions to run a program.

Consist of.

Related to:

- * memory access
- * Instructions is read from memory
- * PC is updated

Related to

- * Control unit
- * Decoding of opcode
- * Identifies operation and addressing mode.

Related to

- * ALU operations
- * Memory access (if required)
- * Control flow.

1. fetch stage
CPU fetches the next instructions on from memory using Program Counter (PC)

2. Decode stage.
CPU decodes the instruction to understand what operation to perform.

3. Execute stage.
CPU performs the operation specified in the instruction.

Purpose
Gets the instructions from memory and loads it into the instruction Register.

Purpose.
Prepares the CPU by identifying the operation and data requirements.

Purpose
Completes the operation and stores the result in register / memory.

After execution of the next instruction is defined and the cycle continues until the program ends.

3. Bus Organisation and its Relationship

System Bus:
Provides a communication pathing b/w CPU memory and I/O devices.

Types of Bus Organisation

1. Single Bus Organisation one bus is shared by CPU memory and I/O devices.

Features

- * Simple & low cost.
- * Only one transaction at a time
- * wide contention $\rightarrow$ slower speed.

uses:

used in small and older system.

2. Multi Bus Organisation
Separate buses for address data and control

Features

- * multiple buses operates simultaneously
- * higher data transfer rate.
- * Less contention

uses:

used in general purpose system

3. Hierarchical Bus Organisation
High speed local buses connected to a system bus

features

- * Organised in levels local system, I/O bus
- * high performance
- * Scalable and flexible.

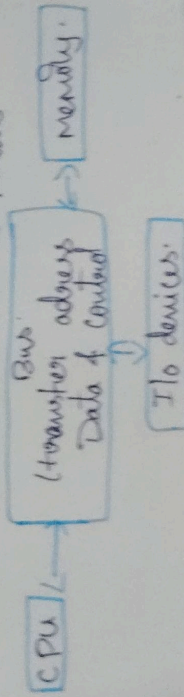
uses:

used in modern high performance computers.

Benefit:

Improves communication efficiency and reduces bottlenecks by managing data transfer effectively.

Relationship with components:



4. 8085 vs 8086 Architecture - Key relationships.

Microprocessor Architecture

Design of CPU, registers, memory & instruction handling.

Consist of

Instruction set:
Set of commands that the processor can execute

8085
8 bit instructions

8086
16 bit instructions

Register Organisation
Selection of storage locations in CPU for fast access.

8085
8 bit registers
LB, CB, D, E, H, P
* Accumulator
* Flag register
* 16 bit PC/SP

8086
* 16 bit general purpose registers
CAX, Bx, CX, DX
* Index registers
(SI, DI)
* Segment registers
(CS, DS, ES)
* IP and flag register

Addressing modes
Method used by instructions to specifying the operands accessed.

8085
Both support
* Immediate
* Register
* Direct
* Implied
* Register Indirect.

Memory Organisation
How memory is structured into segments for better organisation.

8085
* 16 bit address bus
* 4K memory space (0000H-FFFFH)
* Simple internal model

8086
Additional
* Based Indexed
* Based Indexed with displacement

8086
* 20 bit address bus
* 1 MB (00000H-FFFFFH) memory
* Advanced memory handling
* Efficient memory use

8086
* 20 bit address bus
* 1 MB (00000H-FFFFFH) memory
* Advanced memory handling
* Efficient memory use

5. Binary & Hexadecimal number systems in comp

